Zeynep Goktepe

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Pasadena, CA 91126

Education

- **California Institute of Technology**
  B.S. Computer Science / GPA: 4.0 — Pasadena, CA

Experience

- **Cedars-Sinai Medical Center** [] — *February 2026 – Present*
  Researcher — Los Angeles, CA
  - Developed ML pipelines for differentiable neuron modeling using Jaxley and gradient-based optimization.
  - Compared multi-objective optimization (MOO) methods for parameter inference and spike-train prediction.
  - Trained neural networks to predict electrophysiological (eFEL) features from biophysical neuron parameters.
- **Shimojo Psychophysics Laboratory** [] — *March 2025 – Present*
  Researcher — Pasadena, CA
  - Developed Unity-based experimental platform for EEG, ECG, eye-tracking, and behavioral studies with 50+ participants.
  - Built data-processing pipelines for multimodal physiological signals and improved flow-state classification accuracy by 15%.
  - Extracted and analyzed synchronization features from neural and behavioral data for team-flow research.
- **Narb** [] — *August 2025 – December 2025*
  Software Engineering Intern — Remote
  - Developed mobile features including a search history page and login flow, enhancing user experience and onboarding.
  - Implemented authentication and backend services with Clerk and Convex, ensuring secure and scalable account management.
  - Restructured TypeScript monorepo modules to streamline CI/CD processes and improve developer productivity.
- **Babak Hassibi Laboratory** [] — *July 2024 – September 2024*
  Researcher — Pasadena, CA
  - Designed and simulated robust controllers for stochastic systems, reducing instability under disturbances.
  - Evaluated 50+ dynamic system scenarios, demonstrating improved resilience across stochastic and worst-case disturbances.
  - Presented research outcomes at the FSRI banquet to faculty and peers.

Projects

- **Autoguidance for Diffusion-Based Inverse Problems** — *PyTorch, Diffusion Models, Bayesian Inference* []
  - Implemented and evaluated autoguidance-based posterior correction methods for DPS, DDNM+, and PnPDM diffusion samplers.
  - Built Gaussian-mixture inverse problem benchmarks with analytically known posteriors to measure posterior fidelity.
  - Ran large-scale experiments (25 replicates, 4,000 samples/run) and analyzed Wasserstein distance, CRPS, and uncertainty calibration metrics.
- **Level Up: Basketball Training App** — *Swift, Firebase, Xcode* [] []
  - Published on Apple App Store with 1,000+ downloads.
  - Built drill library, workout tracker, and Firebase backend for secure data + video storage.
- **Prize-Fetching Autonomous Robot** — *Python, ROS2, Multithreading* []
  - Implemented ROS2 multi-threaded robot with IR, NFC, and magnetometer sensors; achieved 95% retrieval rate over 30+ trials.
  - Integrated Dijkstra's algorithm with real-time sensor fusion for autonomous navigation.

Skills

- **Programming Languages:** Python, C++, Java, C#, Swift, TypeScript, HTML/CSS
- **Frameworks/Tools:** ROS2, Unity, Convex, Firebase, Git, NumPy, Pandas, SciPy, Matplotlib

Additional Information

Clubs: Society of Women Engineers (SWE), AI Alignment, Sports Analytics
Languages: English (Bilingual), Turkish (Bilingual)
Interests: Basketball, Guitar